

Ngspice: Spice Circuit Simulator

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Ngspice is an open source Spice circuit simulator with its code based on three open source software packages, namely, Cider1b1, Xspice and Spice3f5. This general-purpose mixed-signal or mixed-level simulation tool works for both linear and non-linear circuit analysis; it can work with circuits containing both active and passive components.

Successor to Spice, Cider and Xspice

Although Spice3 is the most popular circuit simulator, Cider provides better simulation accuracy and Xspice extends the code-modelling support, Ngspice is considered to be the successor to all three.

Generally, circuit simulators are quite complex, and introducing improvements in such software is a tough task. However, Ngspice (the project) tries to bring these improvements by fixing bugs, refactoring codes and creating new features in Ngspice (the software), making it the improved version of Spice3f5.

Ngspice and Xspice

Xspice supports Ngspice in many ways. Xspice codes integrated into Ngspice help it to support both mixed-signal and board-level simulation. You may ask: How? The answer to this is very simple. Xspice codes in Ngspice allow it to inherit Xspice framework, which is a mixed-mode simulator that supports analogue as well as digital algorithms for circuit simulation. This ability also makes Ngspice a native mode simulator with an ability to provide two types of simulation in the same executable.

Additionally, Xspice integrates C language code models into Ngspice,

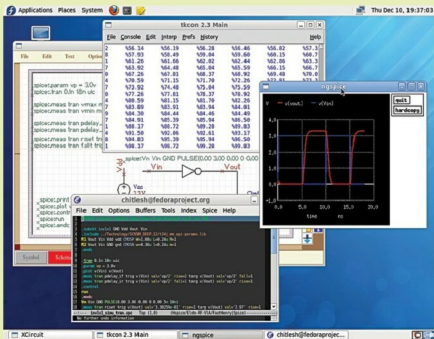


Fig. 1: Ngspice working on Fedora



Fig. 2: Ngspice netlist

Competitive tools

➤ Hspice ➤ PSpice ➤ Spectra

which supports behavioural modelling by various devices like oscillators, filters and amplifiers.

Features

Developed by Paolo Nenzi and his team, Ngspice comes with many features that enable us to consider it

Overview

Version: 26plus (Ngspice rework)

License: Berkeley Software Distribution (BSD) licence

Operating system: LINUX, MS Windows, MAC, BSD, Solaris, etc

Key features:

1. Supports multiple circuit analysis
2. Can be compiled as a shared library, offering full simulator control over a calling process